

A: Datasheet

Algorithm: megvii_005

Developer: Megvii/Face++

Submission Date: 2024_02_20

Template size: 2049 bytes

Template time (2.5 percentile): 1347 msec

Template time (median): 1349 msec

Template time (97.5 percentile): 1360 msec

Investigation:

Frontal mugshot ranking 68 (out of 481) -- FNIR(1600000, 0, 1) = 0.0012 vs. lowest 0.0008 from intema_001

Mugshot webcam ranking 59 (out of 443) -- FNIR(1600000, 0, 1) = 0.0078 vs. lowest 0.0054 from sensetime_009

Mugshot profile ranking 12 (out of 412) -- FNIR(1600000, 0, 1) = 0.0547 vs. lowest 0.0510 from nec_009

Immigration visa-border ranking 10 (out of 370) -- FNIR(1600000, 0, 1) = 0.0008 vs. lowest 0.0005 from psl_001

Immigration visa-kiosk ranking 2 (out of 314) -- FNIR(1600000, 0, 1) = 0.0394 vs. lowest 0.0387 from cloudwalk_mt_002

Identification:

Frontal mugshot ranking 14 (out of 481) -- FNIR(1600000, T, L+1) = 0.0012, FPIR=0.003000 vs. lowest 0.0010 from sensetime_009

Mugshot webcam ranking 11 (out of 441) -- FNIR(1600000, T, L+1) = 0.0084, FPIR=0.003000 vs. lowest 0.0065 from sensetime_009

Mugshot profile ranking 6 (out of 411) -- FNIR(1600000, T, L+1) = 0.0771, FPIR=0.003000 vs. lowest 0.0591 from cloudwalk_mt_002

Immigration visa-border ranking 12 (out of 369) -- FNIR(1600000, T, L+1) = 0.0018, FPIR=0.003000 vs. lowest 0.0009 from cloudwalk_mt_002

Immigration visa-kiosk ranking 35 (out of 314) -- FNIR(1600000, T, L+1) = 0.0738, FPIR=0.003000 vs. lowest 0.0460 from nec_009

B: Mugshot natural images, identification mode: FNIR(N, L+1, T) vs. most accurate (nec_009)

False negative identification rate, FNIR(N)

1e+06

3e+06

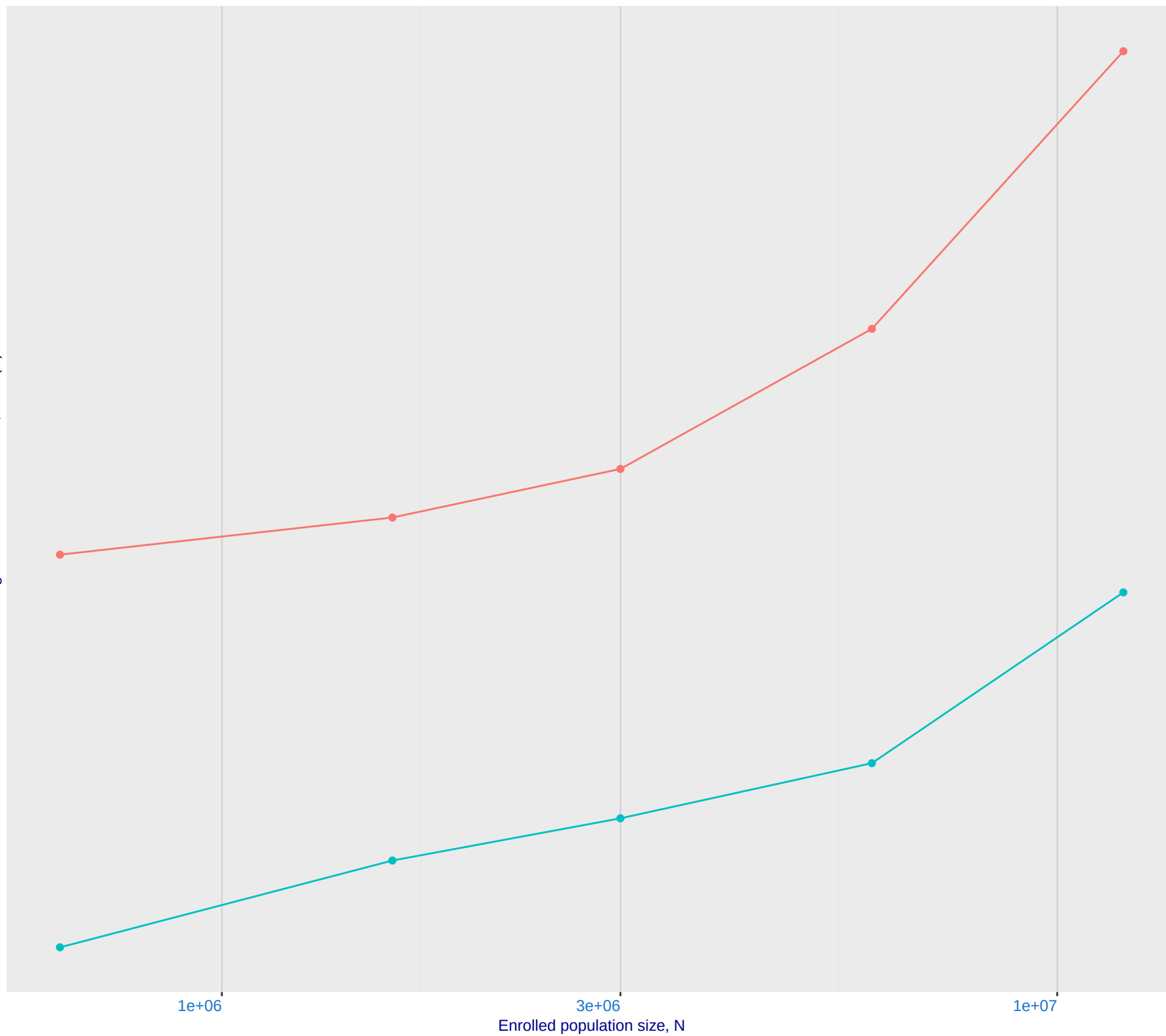
1e+07

Enrolled population size, N

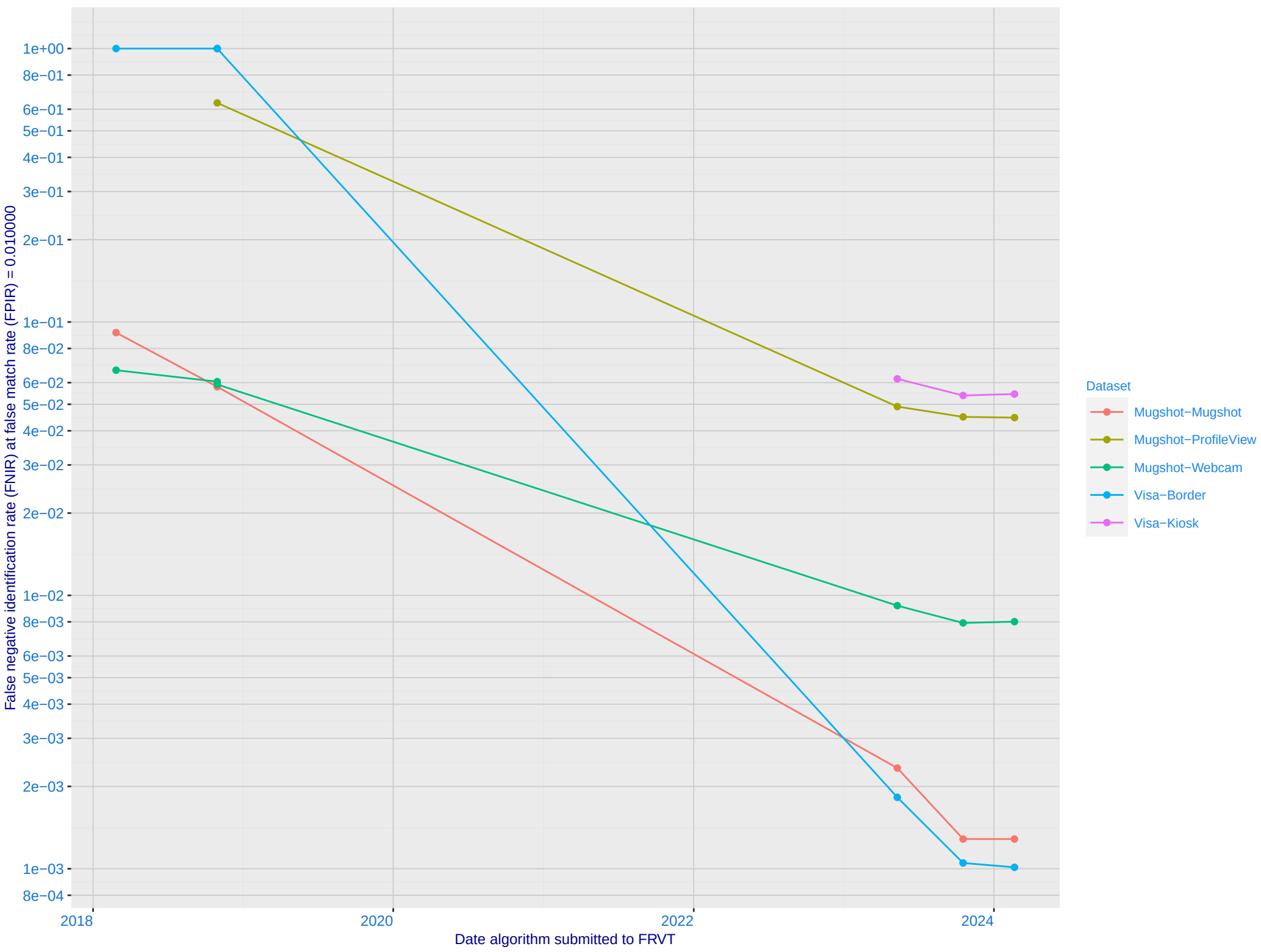
Dataset:
2018 Mugshot
FNIR@FPIR = 0.003

megvii_005
nec_009

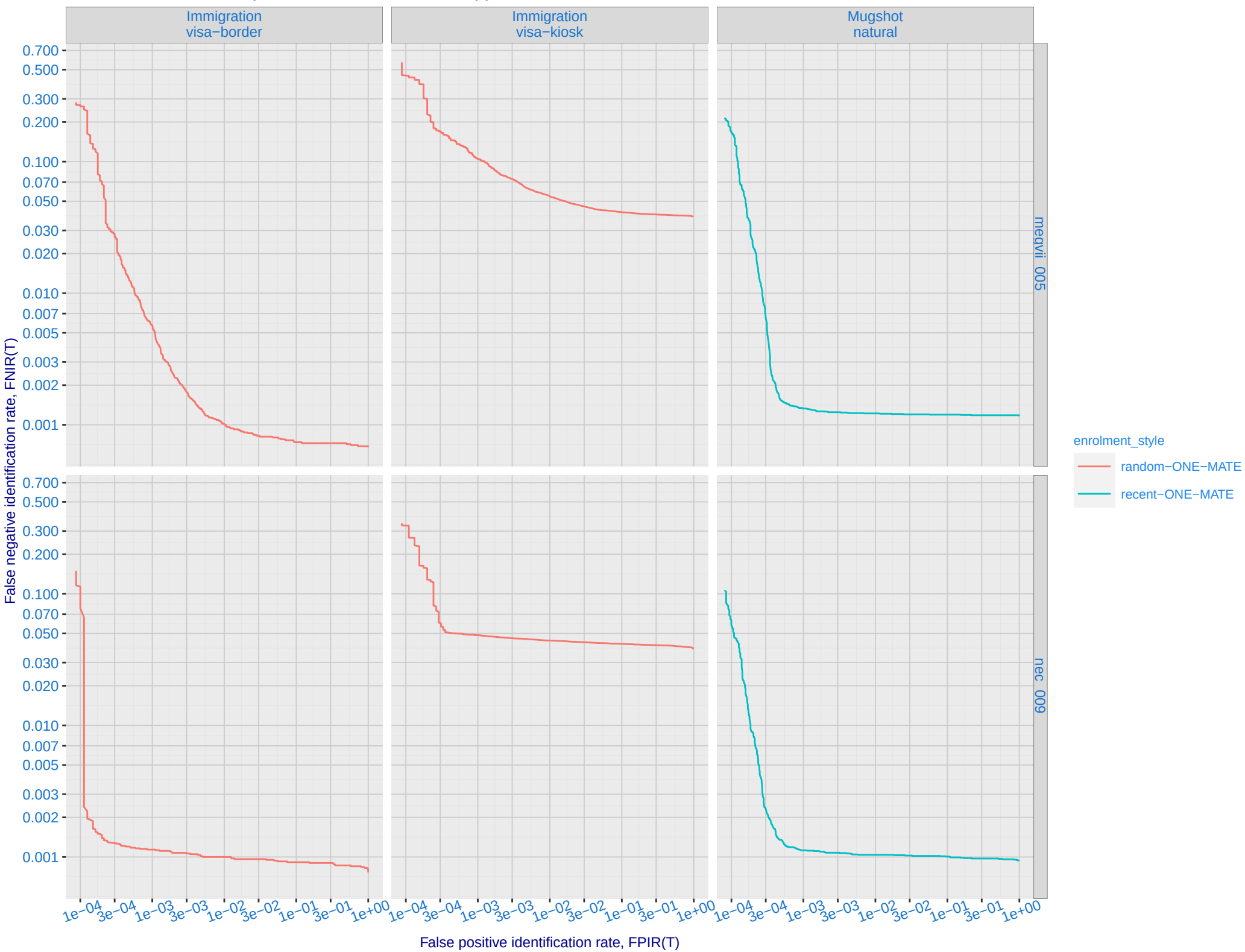
enrolment_style
recent



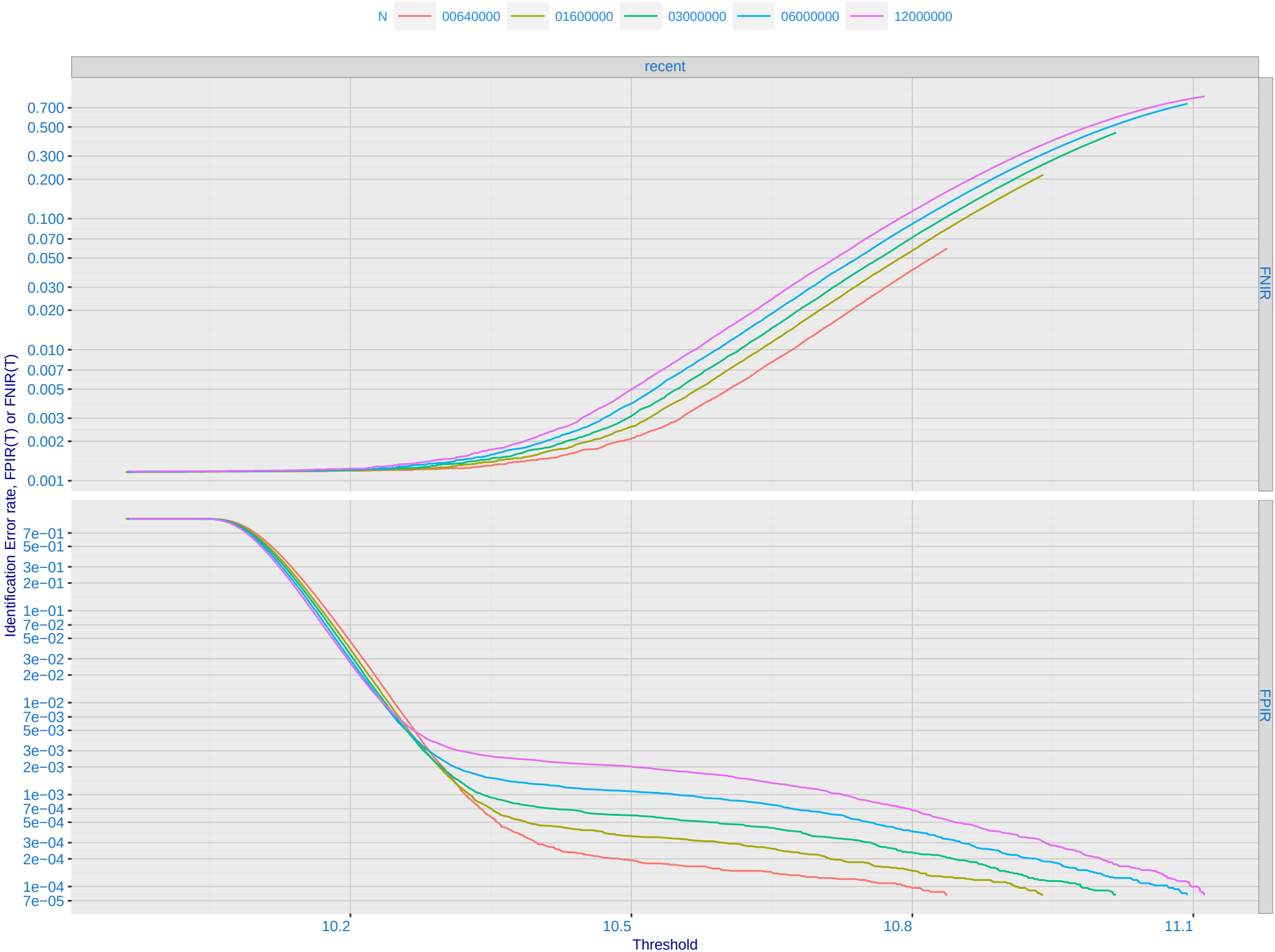
C: Evolution of accuracy for MEGVII algorithms on three datasets 2018 – present



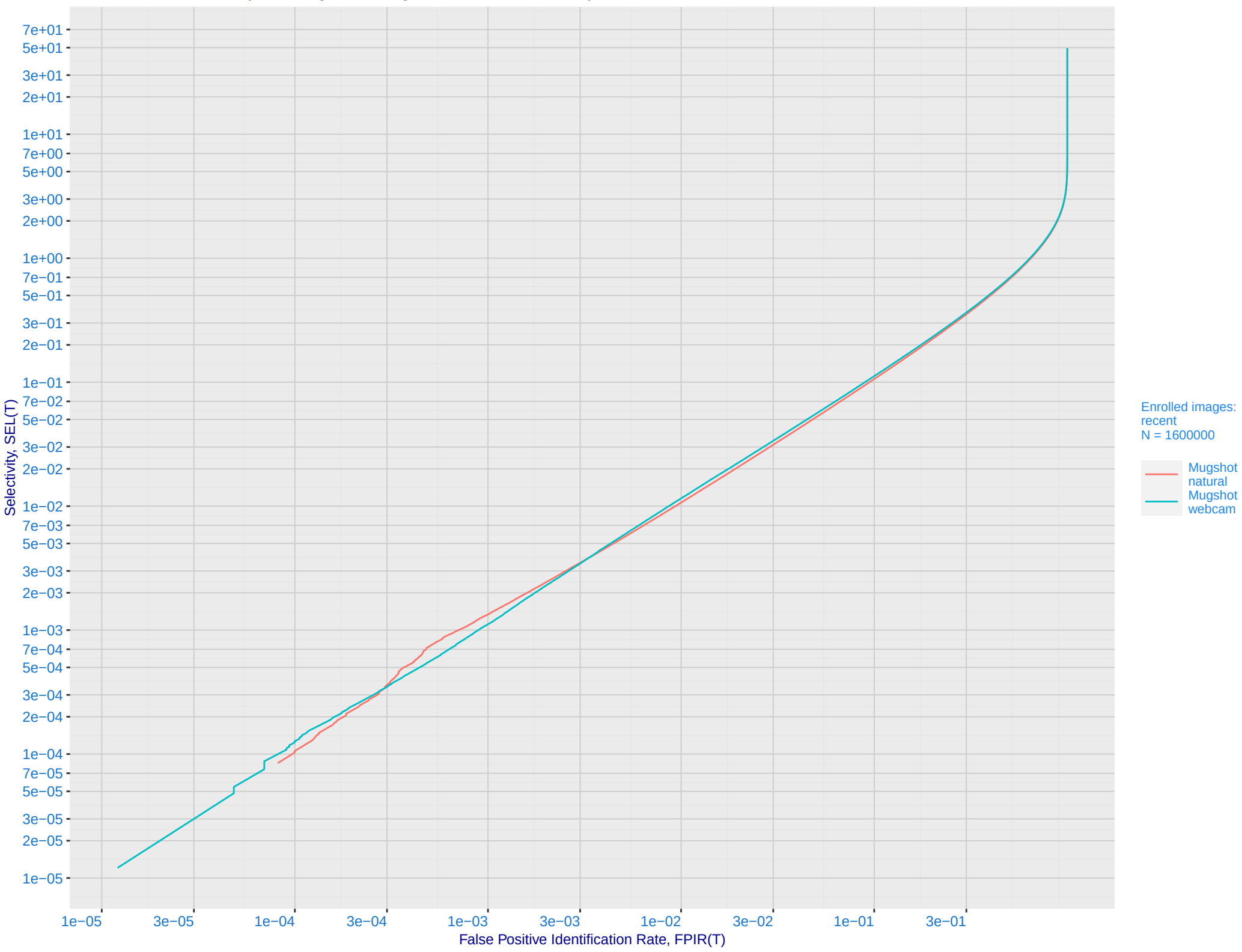
D: 1:N error tradeoff by dataset and enrollment type. N = 1600000 individuals



E: Dependence of error rates on T by number enrolled identities, N, for Mugshot natural images



F: FPIR vs. Selectivity for mugshot images, N = 1600000 subjects enrolled with one recent mate



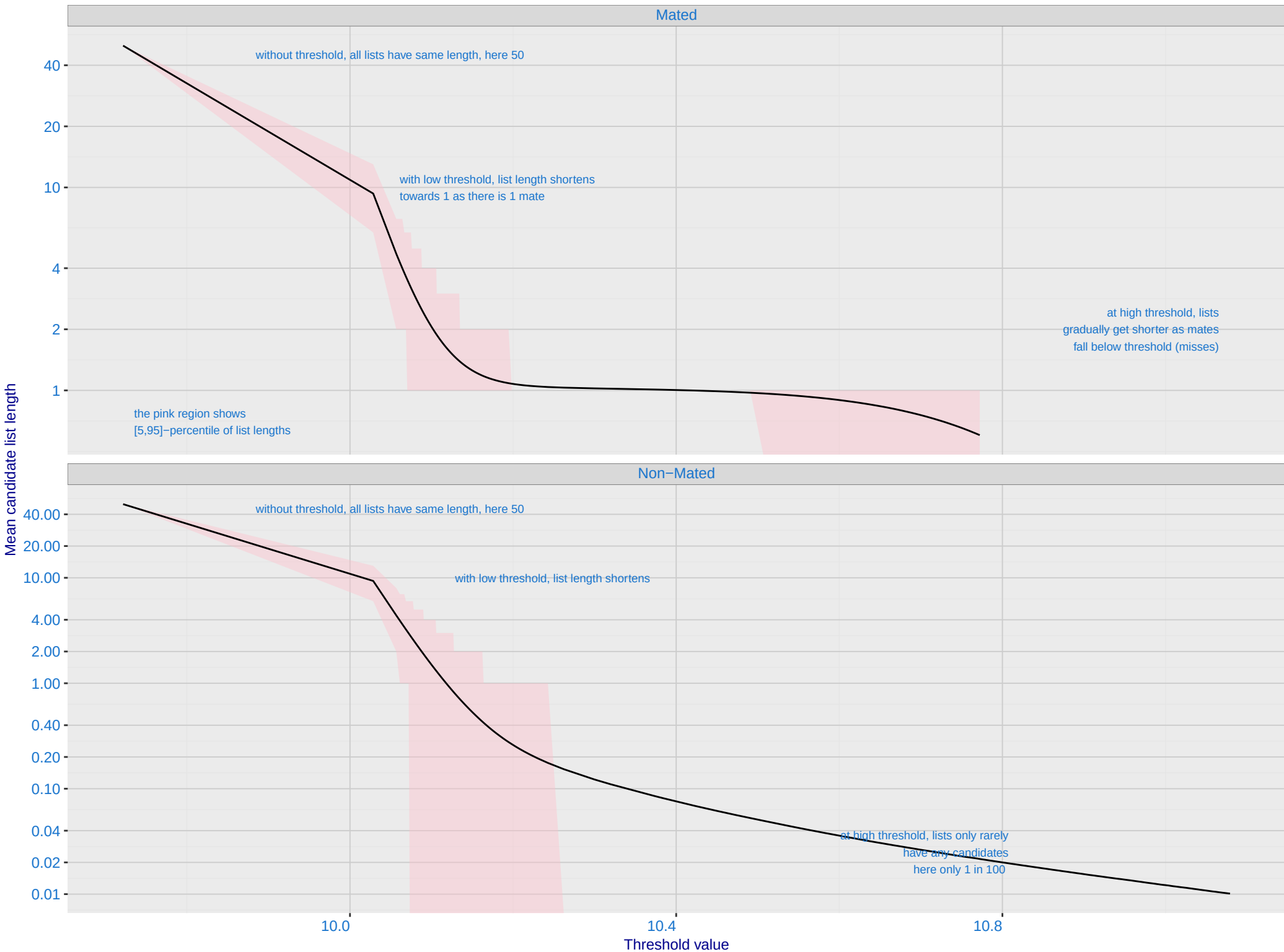
- Immigration visa-border
- Immigration visa-kiosk
- Mugshot
- natural
- Mugshot profile
- Mugshot webcam

enrolment_style

—	random
- - - - -	recent

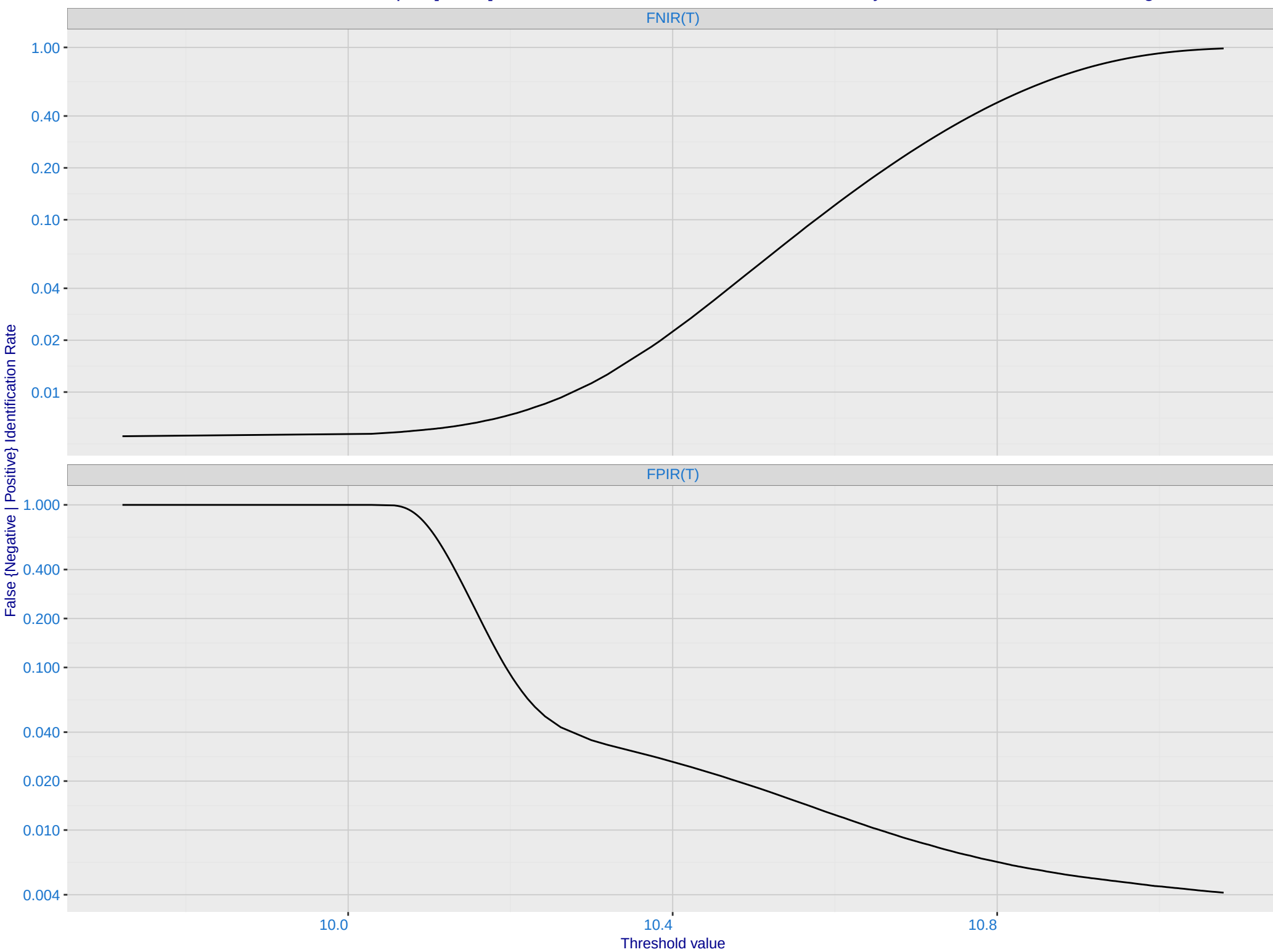
H: Reduced length candidate lists for human review

Dataset is border–border with time–lapse [10,15] YRS with N = 1600000. Probes are 10–15 years later than enrollment image

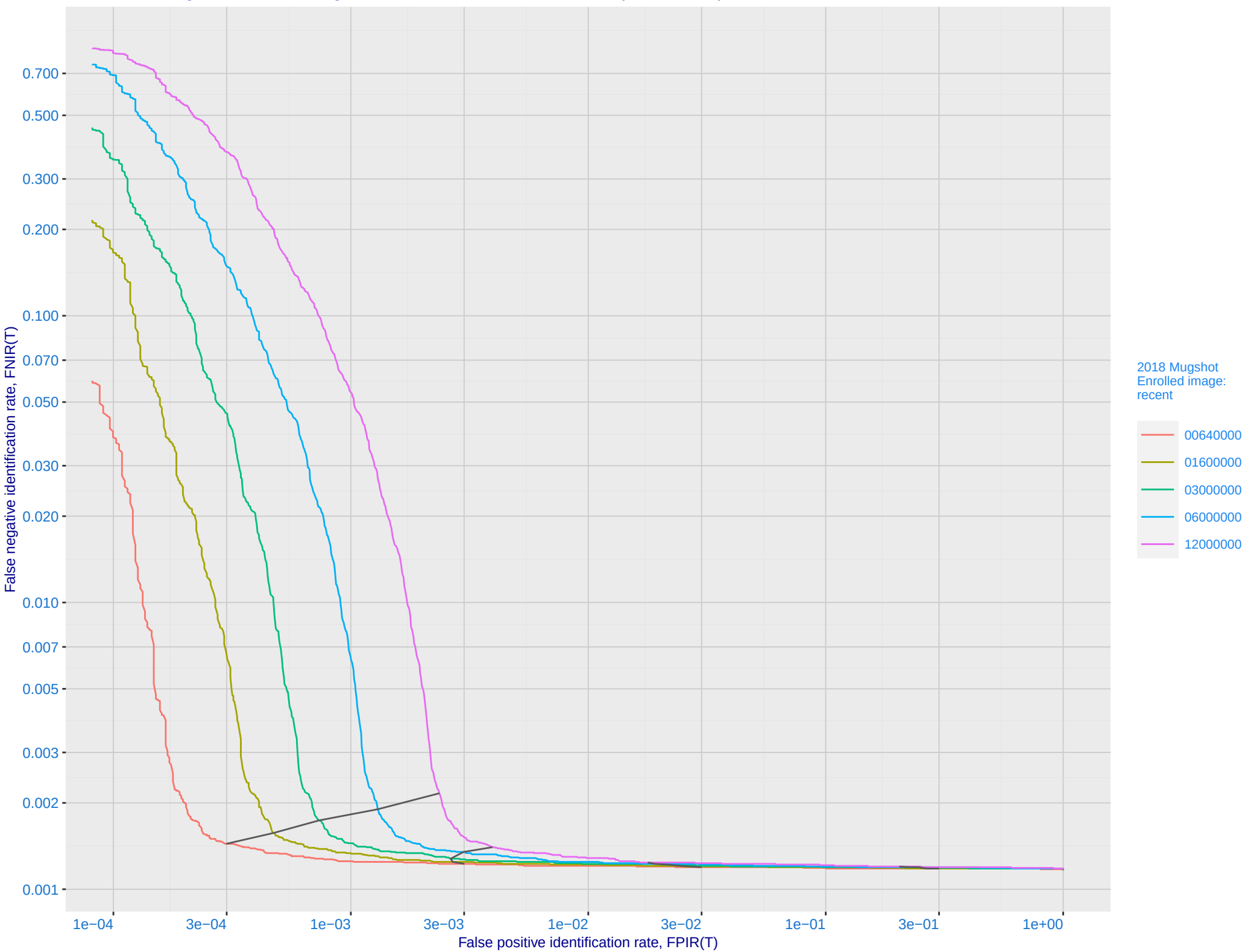


I: FNIR and FPIR dependence on threshold

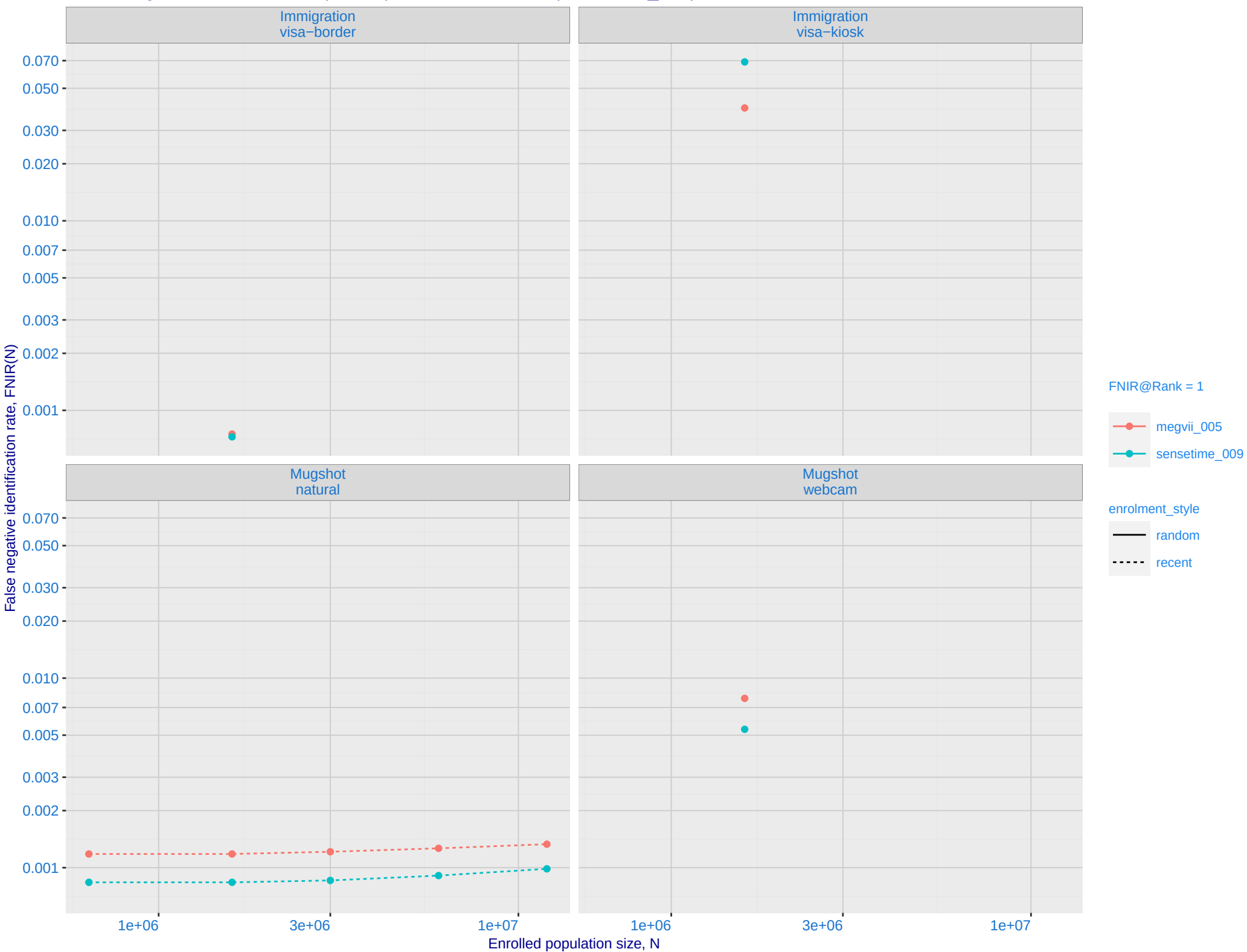
Dataset is border-border with time-lapse [10,15] YRS with N = 1600000. Probes are 10-15 years later than enrollment image



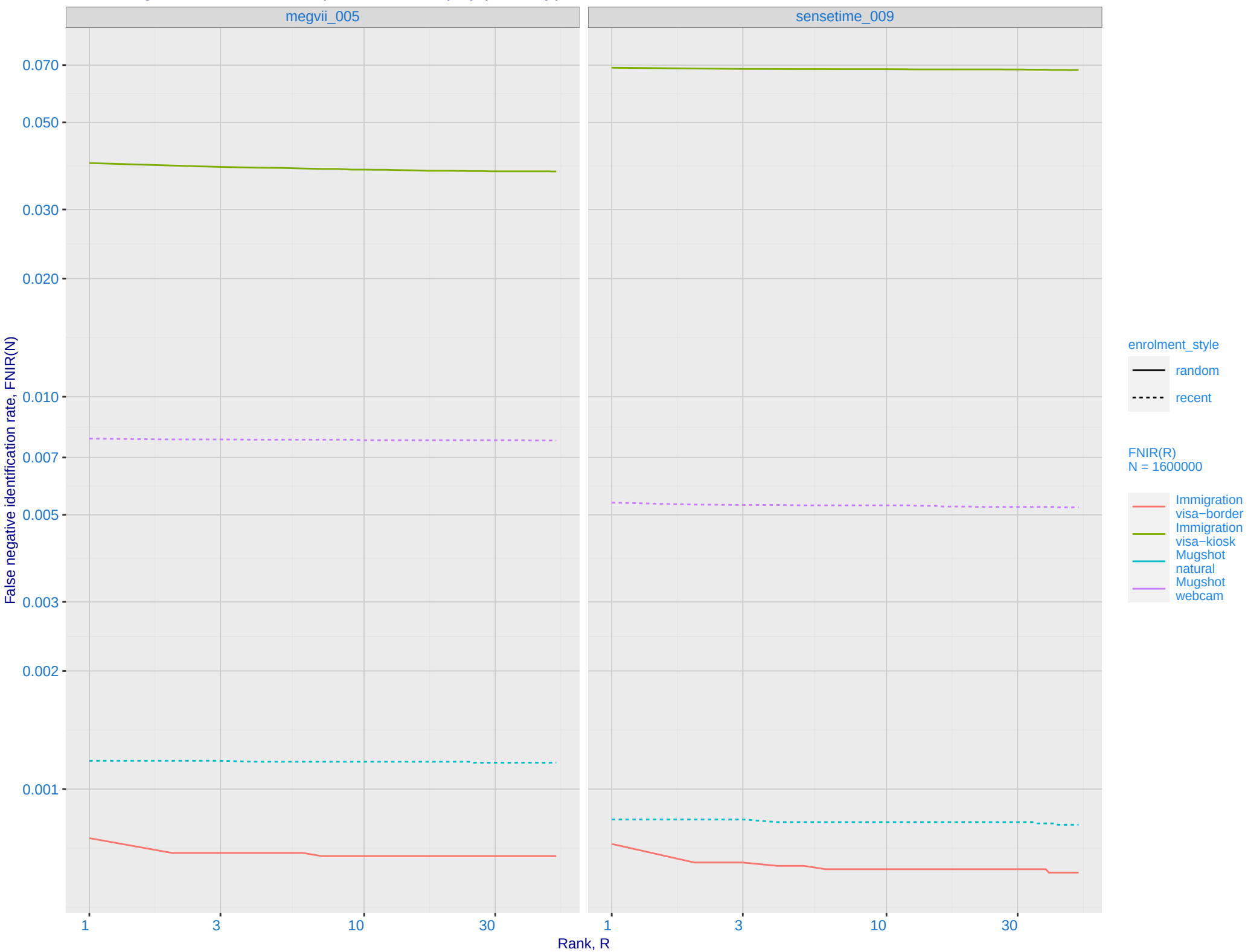
J: DET for Mugshot natural images and various N. Links connect points of equal threshold.



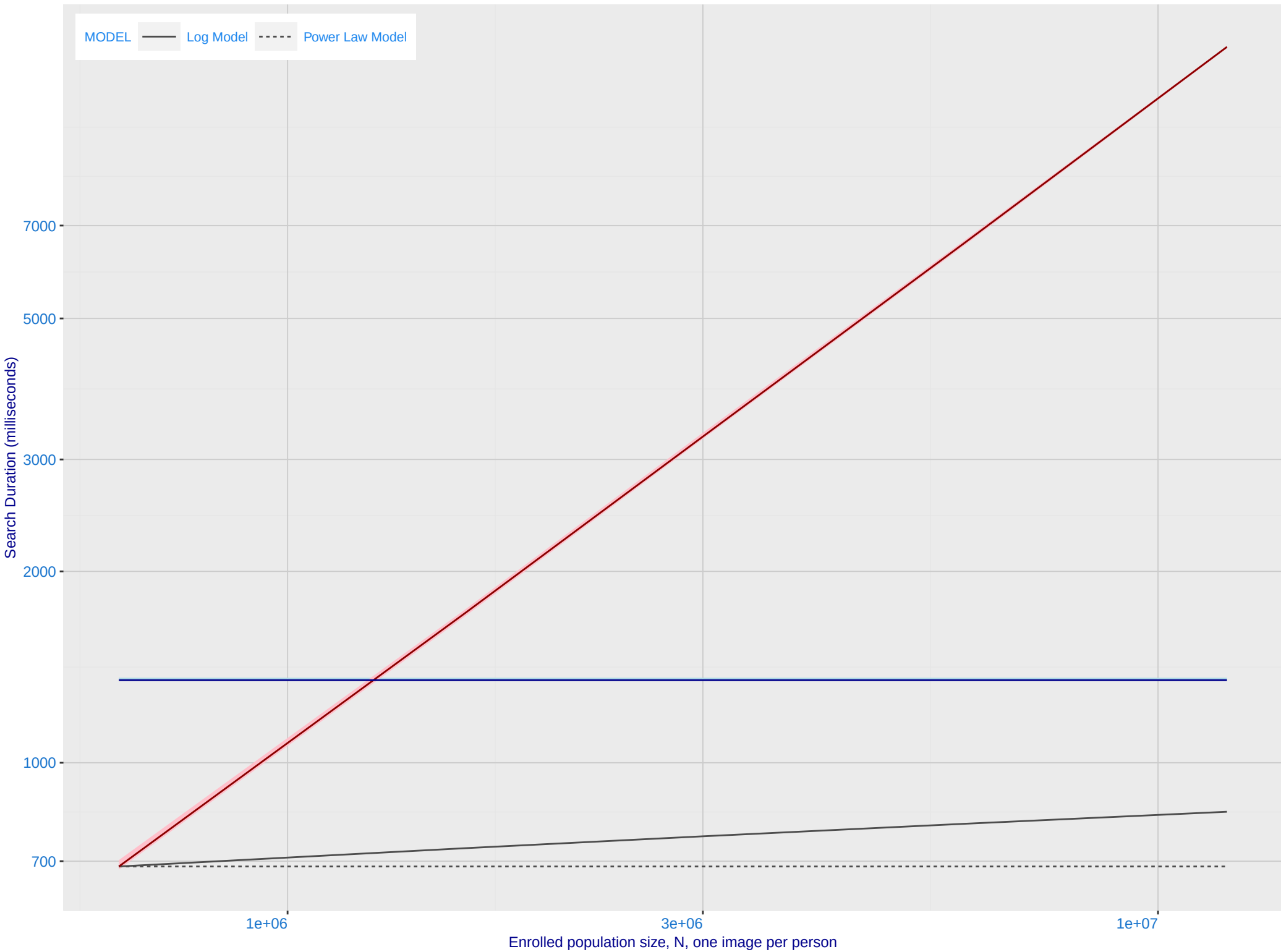
K: Investigational mode: FNIR(N, 1, 0) vs. most accurate (sensetime_009)



L: Investigational mode: FNIR(1600000, R, 0) by probe type

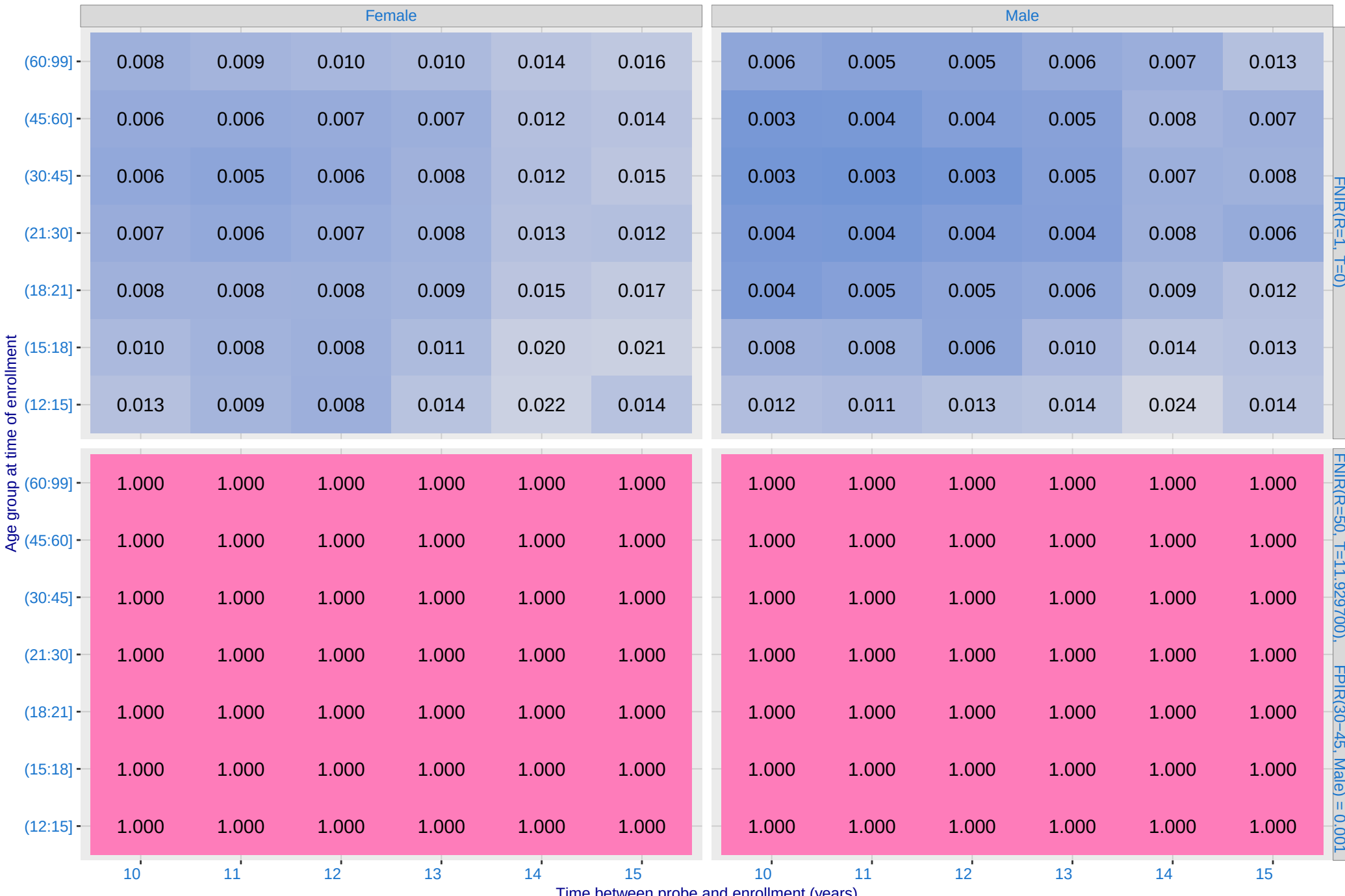


M: Template duration; search duration vs. N. The blue and pink ribbon covers 95 percent of observed measurements. The template generation time is independent of N. The log and power-law models are fit to the first two (N,T) observations



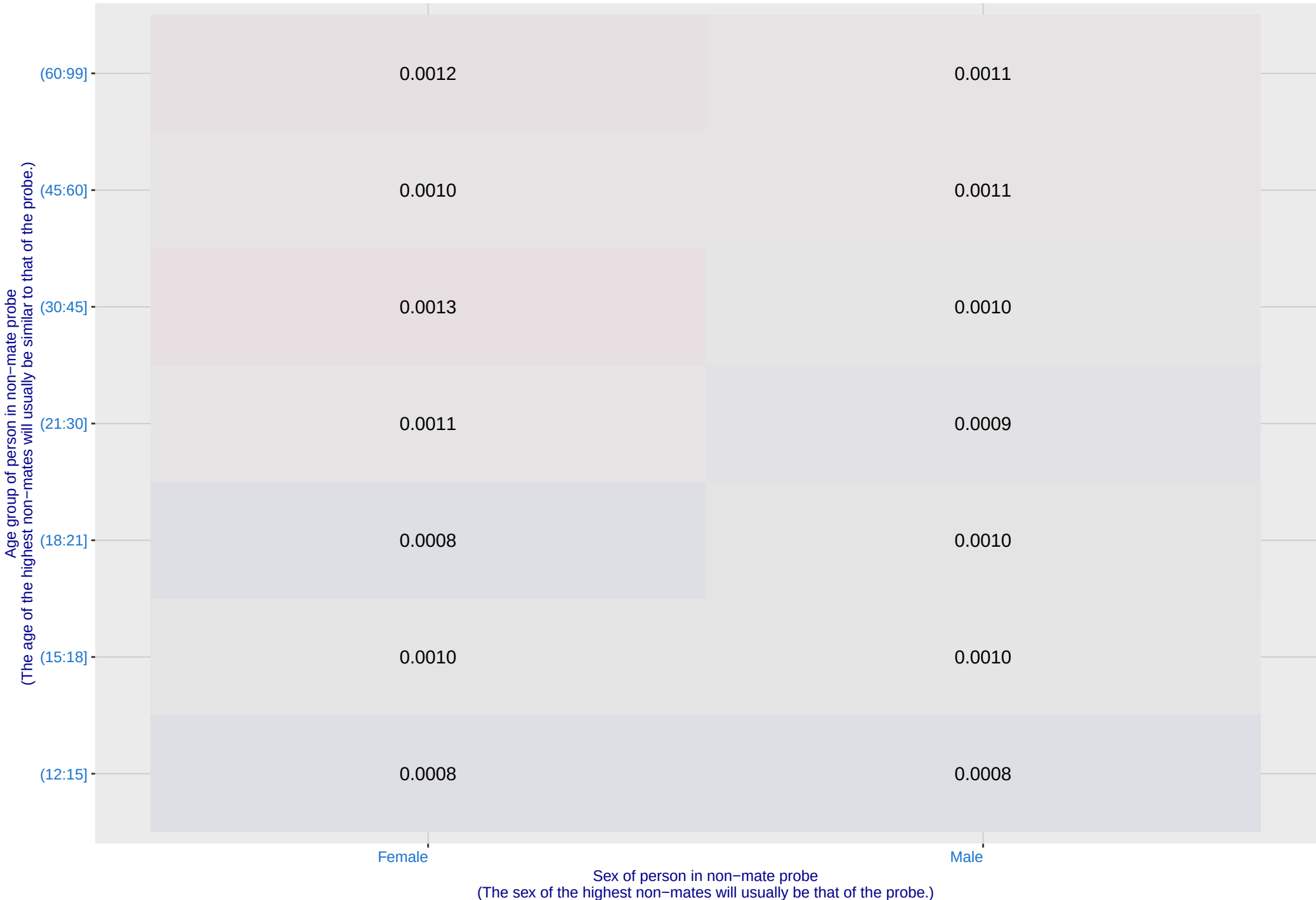
O: FNIR(T, N = 1.6 million) by sex, age and time-lapse. The top row gives investigational rank-1 miss rates. The bottom panels give high threshold for more lights-out identification with low FPIR.

Algorithm: megvii_005, Dataset: Border-Crossing Ageing N = 1600000
Text encodes FNIR, Color encodes log(FNIR)



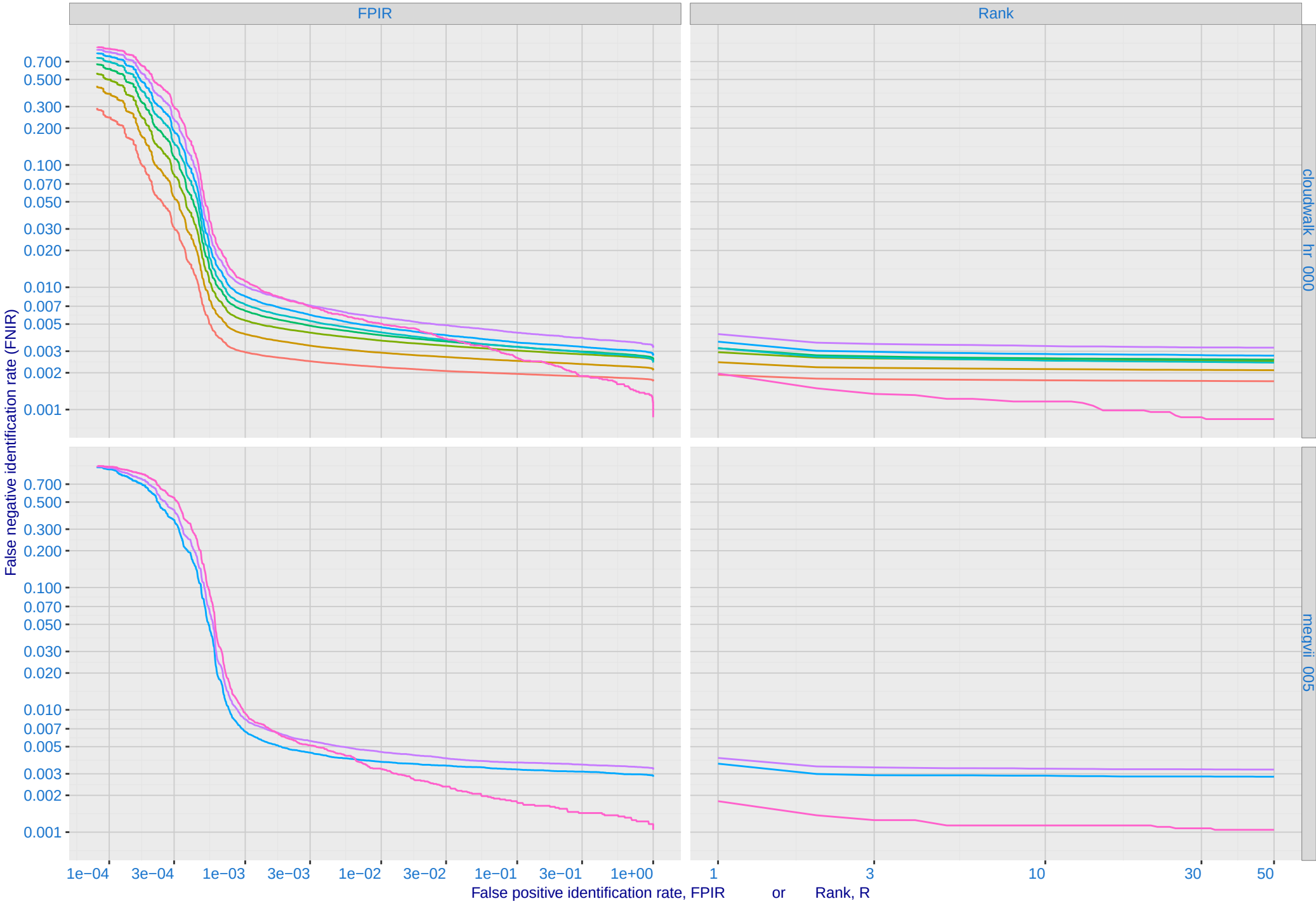
P: FPIR(N = 1.6 million) by sex and age. It is typical for false positive identification rates to be higher in women except in their teens.

Algorithm: megvii_005, Dataset: Border-Crossing Ageing
Threshold: 11.929700 set to achive FPIR(30-45, Male) = 0.001



Q: Identification FNIR(N, T, L+1) and Investigational FNIR(N, 0, R) under ageing

Dataset: 2018 Mugshot N = 3068801



R: Decline of genuine scores with ageing, with some eventually dropping below typical thresholds shown by the horizontal lines

